MECHTRON / SFWRENG 3K04: Software Development
Project Overview – Fall 2026

McMaster University

Introduction

Welcome to MT/SE 3K04! For the course project, you will get to develop an automated dispensing cabinet (ADC) and medicine monitoring system used in hospitals. We would like to give you some helpful information on what to expect and how to complete the course project successfully.

The project will be conducted in teams of 4 to 5. All team members must be in the same lab section. During your first lab, if you do not have a full team, your TAs will assist you with finding a match with other students who are also looking to build a team. Labs will begin in the second week of the semester and will be held every week (with the exception of the mid semester break week and the last week of classes). During the labs, you will have designated time to consult your TAs, and to continue work on the project with your group.

The course project is divided into **three deliverables**. Each deliverable has two components: (1) a **technical demonstration** with the TAs during your lab, and (2) your deliverable **submission files**. The project deliverable schedule and deadlines can be found in the Lab Schedule section of this document.

The first deliverable will be focused on requirements and scope, the second deliverable will be focused on the design and implementation, and the third one will be around safety and integration. For the first two deliverables, as your team will be split into two sub-groups working on two different parts of the project.

- **Part A - Automated Dispensing Cabinet (ADC):** The embedded software that controls a secure, software-driven cabinet at the point of care. It handles user authentication, dispensing, inventory, event logging, fault detection, operational modes, and communication with an external monitor.
- **Part B - Hospital Monitoring System:** The supervisory, human-facing application that talks to one or more ADCs. It handles ADC communication, inventory supervision, alarm management, historical data, reporting, dashboards, and cabinet-health monitoring.

The two sub-groups will be marked and assessed separately. That is, for the first two deliverables, your team sub-groups may receive different grades. For the last deliverable, everyone will be marked and assessed together. However, we still expect you to collaborate and make decisions together throughout the project, as the decisions you make during the first deliverable are very important.

If you find that getting started on the project seems overwhelming at first, don't despair! You will get the hang of it. You will have lots of support from your TAs along the way. Do not hesitate to ask us if you have any questions during the lab or Piazza. We have also prepared a best practices document with some software development recommendations. We hope that the course project will be a valuable learning experience for you.

— 3K04 Teaching Team

Team Contracts

For a successful software development project, you need to get your expectations straight as a team. This includes agreeing on communication means and frequency, accountability, group policies, etc.

Therefore, we are sharing two different templates of a *Team Contract*. Together with your team, you will choose one of those contracts to discuss and complete during the first lab. It is important that you take those decisions seriously, and use it as a binded document to address any team conflict.

ReQuest Tool

A pretty important skill that we wish we could put more emphasis in, is learning how to ask questions. In a computing and engineering context, that skill is useful in settings such as elliciting requirements. That is why, we are introducing a new educational tool called *ReQuest*, to help you practicing those skills. The tool will help you clarify some of the requirements for the projects and give you feedback on your question asking skills.

For this project, you must use the tool at least once during the semester. While it will be more helpful to use it at the beginning of the semester for Deliverable 1, you can use it at any point during the project. More details will be available on Avenue, with instructions on how to access the tool.

Evaluation

The project is 35% of your overall grade. Find below how the 35% is broken down.

Group Contract	1%
Use of the ReQuest Tool	2%
Deliverable 1	10%
Deliverable 2	10%
Deliverable 3	12%
Total	35%

Lab Schedule

This schedule covers what you must work on during the lab. While there isn't a particular timeline for your project (other than the deadlines), it is your responsibility to make constant progress on it, and use the lab as a set time to work as a team and get help from the TAs.

- **Week of Sep. 14th:** First week of labs!
  - Form your teams and inform your TAs.
  - Complete and sign your team contracts and upload them to Avenue by Friday, September 18th at 11:59 PM.

- **Week of Sep. 21st:** Getting help from the TAs on the project.
- **Week of Sep. 28th:** Getting help from the TAs on the project.
- **Week of Oct. 5th:** Deliverable 1 demo week!
  - Technical demonstration to your TAs.
  - Submit your Deliverable 1 by Friday, October 9th at 11:59 PM.
- **Week of Oct. 12th:** Break week, enjoy pumpkin and/or apple pie.
- **Week of Oct. 19th:** Getting help from the TAs on the project.
- **Week of Oct. 26th** Getting help from the TAs on the project.
- **Week of Nov. 2nd:** Deliverable 2 demo week!
  - Technical demonstration to your TAs.
  - Submit your Deliverable 2 by Friday, November 6th at 11:59 PM.
- **Week of Nov. 9th:** Getting help from the TAs on the project.
- **Week of Nov. 16th:** Getting help from the TAs on the project.
- **Week of Nov. 23rd:** Getting help from the TAs on the project.
- **Week of Nov. 30th:** Deliverable 3 demo week!
  - Technical demonstration to your TAs.
  - Submit your Deliverable 3 by Thursday, December 3th at 11:59 PM.